2.3 **[1.0점]** $|\Delta U_S|$ $\Delta U_G$ ,

. .

$$\sigma_0 \cos\theta \cdot 2\pi R \Delta h = \pi R^2 \Delta h \rho g h \quad \Rightarrow \quad h_0 = \frac{2\sigma_0 \cos\theta}{\rho g R}. \tag{6}$$

$$h_0 = \frac{2\sigma_0 \cos\theta}{\rho g R} = \frac{2 \cdot 0{,}072 \cdot \cos 20°}{1{,}0 \cdot 10^3 \cdot 9{,}8 \cdot 1{,}0 \cdot 10^{-3}} = 1{,}4 \cdot 10^{-2} m = 14 mm. \tag{7}$$

## 3. 수직 원뿔형 관 안의 물 (4.0점)

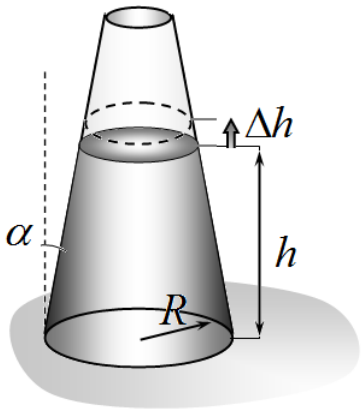
3.1 **[0.5점]** .

$\Delta U_S$

$\Delta h$

$$\Delta U_S = -\sigma_0 \cos\theta \cdot \Delta S = -\sigma_0 \cos\theta \cdot 2\pi r \frac{\Delta h}{\cos\alpha}. \tag{8}$$

$r = R - h\,tg\alpha$ , $h$ .




3.2 **[0.5점]**

$\Delta U_G$

$\Delta h$

$$\Delta U_G = \pi r^2 \Delta h \rho g h. \tag{9}$$

3.3 **[1.0점]** ,

$$\sigma_0 \cos\theta \cdot 2\pi r \frac{\Delta h}{\cos\alpha} = \pi r^2 \Delta h \rho g h, \tag{10}$$

$h$

$$\frac{2\sigma_0 \cos\theta}{R - h\,tg\alpha} = \rho g h \cos\alpha, \tag{11}$$

$h_0$

$$\frac{2\sigma_0 \cos\theta}{\rho g R\left(1 - \frac{h}{R} tg\alpha\right)} = h\cos\alpha \quad \Rightarrow \quad \frac{h_0}{1 - \frac{h}{R} tg\alpha} = h\cos\alpha, \tag{12}$$

3.4 **[1.0점]** $h$ .

. (12)

$$h_0 = h\cos\alpha\left(1 - \frac{h}{R} tg\alpha\right). \tag{13}$$

$h = 0$ $h = \frac{R}{tg\alpha}$ .

$\frac{R}{4tg\alpha}\cos\alpha$ , $h = \frac{R}{2tg\alpha}$

. (12)

$h_0 > \frac{R}{4tg\alpha}\cos\alpha$ . .

$$\frac{R}{4tg\alpha}\cos\alpha = 25mm$$

.

